REDUCED CLOSED SUBSCHEMES OF A NODAL CURVE ARE NODAL, AND MEET THEIR COMPLEMENT IN FINITE SEPARABLE SPECTRA AT NODES OF X

THE STACKS PROJECT AUTHORS

Editorial note. All delivered mathematical text is the author's own, copied from the pinned Stacks Project chapter file curves.tex; no statement or proof step was rewritten, reordered or removed.

Editorial note. The counted proof is the author's complete argument: the component-union reduction, the full case analysis including the two-branch situation, the completed uv/(uv) computation of the intersection ideal, and the closing regularity-plus-separability smoothness argument are all retained verbatim.

Editorial note. The label lemma-closed-subscheme-nodal-curve names the counted statement, and the proof environment immediately following it in the source is delivered verbatim as its proof.

Editorial note. The counted statement is carried by the author's own lemma environment in the chapter source.

Allowed standard prerequisites (editorial).

- Components through a node. Properties, Lemma properties-lemma-number-of-branches-irreducible-components; Same chapter, Lemma lemma-nodal. Bounds the number of irreducible components through a point by the number of geometric branches, which is 2 at a node.
- The completed uv/(uv) normal form at a node. Same chapter, Remark remark-trivial-quadratic-extension and Lemma lemma-nodal; More on Algebra, Lemma more-algebra-lemma-completion-regular; Algebra, Lemma algebra-lemma-separable-smooth. Identifies the completed local ring as kappa(y)[[u, v]]/(uv), makes the completed maps to Y and Z the two coordinate projections, and upgrades regularity plus separability to smoothness.

Source reference. Original lemma 00TV: sources/stacks-chow/tags.txt, source lines 1089–1089. Prerequisite pointer invoked at proof line(s) 4858; it names a standard fact of a sibling chapter that the counted argument uses, and every step of the counted proof built on it is retained verbatim in the delivered body.

Source reference. Original reference 0C4D: sources/stacks-curves/pinned-original-source.tex, source lines 4391–4391. Prerequisite pointer invoked at proof line(s) 4826; it names a standard fact of the same chapter that the counted argument uses, and every step of the counted proof built on it is retained verbatim in the delivered body.

Source reference. Original lemma 07NY: sources/stacks-chow/tags.txt, source lines 9778–9778. Prerequisite pointer invoked at proof line(s) 4855; it names a standard fact of a sibling chapter that the counted argument uses, and every step of the counted proof built on it is retained verbatim in the delivered body.

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Source reference. Original lemma 0E20: sources/stacks-chow/tags.txt, source lines 17590–17590. Prerequisite pointer invoked at proof line(s) 4829; it names a standard fact of a sibling chapter that the counted argument uses, and every step of the counted proof built on it is retained verbatim in the delivered body.

Source reference. Original reference 0CBU: sources/stacks-curves/pinned-original-source.tex, source lines 4549–4549. Prerequisite pointer invoked at proof line(s) 4840; it names a standard fact of the same chapter that the counted argument uses, and every step of the counted proof built on it is retained verbatim in the delivered body.

Lemma 0.1. *Let k be a field. Let X be a locally algebraic k-scheme equidimensional of dimension 1 whose singularities are at-worst-nodal. If $Y \subset X$ is a reduced closed subscheme equidimensional of dimension 1, then*

(1) *the singularities of Y are at-worst-nodal, and*
(2) *if $Z \subset X$ is the scheme theoretic closure of $X \setminus Y$, then*
   (a) *the scheme theoretic intersection $Y \cap Z$ is the disjoint union of spectra of finite separable extensions of k,*
   (b) *each point of $Y \cap Z$ is a node of X, and*
   (c) *$Y \to \mathrm{Spec}(k)$ is smooth at every point of $Y \cap Z$.*

Proof. Since X and Y are reduced and equidimensional of dimension 1, we see that Y is the scheme theoretic union of a subset of the irreducible components of X (in a reduced ring (0) is the intersection of the minimal primes). Let $y \in Y$ be a closed point. If y is in the smooth locus of $X \to \mathrm{Spec}(k)$, then y is on a unique irreducible component of X and we see that Y and X agree in an open neighbourhood of y. Hence $Y \to \mathrm{Spec}(k)$ is smooth at y. If y is a node of X but still lies on a unique irreducible component of X, then y is a node on Y by the same argument. Suppose that y lies on more than 1 irreducible component of X. Since the number of geometric branches of X at y is 2 by Lemma 0C4D, there can be at most 2 irreducible components passing through y by Properties, Lemma 0E20. If Y contains both of these, then again $Y = X$ in an open neighbourhood of y and y is a node of Y. Finally, assume Y contains only one of the irreducible components. After replacing X by an open neighbourhood of x we may assume Y is one of the two irreducble components and Z is the other. By Properties, Lemma 0E20 again we see that X has two branches at y, i.e., the local ring $\mathcal{O}_{X,y}$ has two branches and that these branches come from $\mathcal{O}_{Y,y}$ and $\mathcal{O}_{Z,y}$. Write $\mathcal{O}^\wedge_{X,y} \cong \kappa(y)[[u,v]]/(uv)$ as in Remark 0CBU. The field $\kappa(y)$ is finite separable over k by Lemma 0C4D for example. Thus, after possibly switching the roles of u and v, the completion of the map $\mathcal{O}_{X,y} \to \mathcal{O}_{Y,Y}$ corresponds to $\kappa(y)[[u,v]]/(uv) \to \kappa(y)[[u]]$ and the completion of the map $\mathcal{O}_{X,y} \to \mathcal{O}_{Y,Y}$ corresponds to $\kappa(y)[[u,v]]/(uv) \to \kappa(y)[[v]]$. The scheme theoretic intersection of $Y \cap Z$ is cut out by the sum of their ideas which in the completion is (u, v), i.e., the maximal ideal. Thus (2)(a) and (2)(b) are clear. Finally, (2)(c) holds: the completion of $\mathcal{O}_{Y,y}$ is regular, hence $\mathcal{O}_{Y,y}$ is regular (More on Algebra, Lemma 07NY) and $\kappa(y)/k$ is separable, hence smoothness in an open neighbourhood by Algebra, Lemma 00TV. □